ME 5309 – Mid-Semester Examination: ANSWER SHEET

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Questions attempted: Q1, Q2, Q4

Question 1

(a) (i) $f_x = f / p = 16\text{ mm} / 5\ \mu\text{m} = 16\times10^{-3} / 5\times10^{-6} =$ **3200 px**.

(ii) Sensor width $W_s = 1280 \times 5\ \mu\text{m} = 6.4$ mm. By similar triangles FOV $= W_s \cdot Z / f = 6.4 \times 400 / 16 =$ **160 mm**.

Distance per pixel = FOV / 1280 = 160/1280 = **0.125 mm/px** (chek: $Z/f_x = 400/3200 = 0.125$ mm/px).

(b) LSB $= 5\text{ V}/2^{12} = 5/4096 = 1.2207$ mV $= 1.2207\times10^{-3}$ V.

$\Delta F = \text{LSB}/S$ with $S = 10$ mV/N $= 10$ V/N (taking the mV as V in the denominator): $\Delta F = 1.2207\times10^{-3}/10 =$ **1.22×10^{-4} N**.

(c) Using forward Euler: $x_{k+1} = x_k + T(A\,x_k + B\,u_k)$, so $F = 1 + AT$ and $G = BT$.

$F = 1 + (-2)(0.1) =$ **0.8**, $G = 1 \times 0.1 =$ **0.1**.

$x_{k+1} = 0.8\,x_k + 0.1\,u_k$.

Question 2

(a) Let $x_1 = \theta$ and $x_2 = \dot{\theta}$. $\dot{x}_1 = x_2$; $\dot{x}_2 = (1/J)[\tau + b\,x_2 - mg\ell \sin x_1]$.

Equilibrium: $x_2 = 0$ and $mg\ell \sin \theta_e = \tau_0$. Becuase $-1 \le \sin\theta \le 1$ we need $|\tau_0| \le mg\ell = 2.4525$ N·m.

(b) $\delta\dot{x} = A\,\delta x + B\,\delta\tau$, $A = \partial f/\partial x$, $B = \partial f/\partial\tau$ at $\theta_e = 60°$, $x_{2e} = 0$.

$A = [0,\ 1;\ -(mg\ell/J)\cos\theta_e,\ -b/J] = [0,\ 1;\ -122.6\times0.5,\ -2.5] =$ **[0, 1; −61.31, −2.5]**.

$B = [0;\ 1/J] =$ **[0; 50]**.

(c) At steady state $\dot{\theta} = 0$ and $\ddot{\theta} = 0$.

A PD controler is stabel and its job is to drive $(\theta_d - \theta)$ to zero, so $\theta_{ss} = \theta_d = 90°$. The gravity term only affects the transient. Hence the steady-state error is **0°** and the joint does come to rest at θ_d.

(d) Graviy compensation reduces the steady-state error to about 0.5°, not exactly zero, since the compensation is evaluated at the measured angle and the sensor and model are not perfect.

Poles: unchanged. The linearisation of the gravity term about θ_d is $mg\ell\cos\theta_d$, wich is zero at 90°, so the stiffness stays K_p.

Question 4

(a) Closed loop: $\ddot{\theta} + (a + bk_2)\dot{\theta} + (c + bk_1)\theta = bN_{ff}\,r$, i.e. $s^2 + 4s + 8 = 0$ so the poles are $s = -2 \pm 2j$.

DC gain $= bN_{ff}/(c + bk_1) = N_{ff}/8$. For unity gain $N_{ff} = 1/8 =$ **0.125**.

(b) The actuator is 25% weaker, so the output is also 25% lower: $\theta_{ss} = 0.75$, and the tracking error = **25%**.

(c) Closed-loop: $\ddot{\theta} + (a + bk_2)\dot{\theta} + (c + bk_1)\theta = 0$, so $s^2 + (a + bk_2)s + (c + bk_1) = 0$.

Poles at $-2 \pm 2j$ give $s^2 + 4s + 8$, so $k_2 = 2$ and $k_1 = 2$. The extra pole at $s = -4$ comes from the integrater, so choose $K_I = 4$.

(d) Steady state: $\dot{\xi} = 0$, so $r = \theta_{ss}$; the parameters b and c do not enter, hence zero error for any b and c as long as it is stable.

Failure mode: windup when the actuator saturates. Remidy: reduce K_I to zero (or use a smaller gain) while saturated.